On the Interaction Basis of the Q-LINK Messenger: Lie Algebra Collapses and Isomorphisms Extension 2

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Abstract

The Q-LINK architecture achieves barren plateau mitigation by routing phase information through a residual messenger qubit using R_{xx} collection interactions. This note investigates the structural dependence of the architecture on its chosen interaction basis. By evaluating the iterated Lie brackets of the circuit generators, we prove that replacing the collection basis with R_{zz} strictly fractures the Dynamical Lie Algebra (DLA), decoupling the messenger and erasing the variance advantage. Conversely, an R_{yy} collection basis guarantees an algebraic isomorphism, preserving the full $\mathfrak{su}(2^{n+1})$ expansion. We conclude that the collection basis must strictly fail to commute with the control-basis of the distribution layer.

1 Lie Generators of the Residual Pathway

The Q-LINK residual pathway is constructed from two primary interaction blocks: the collection unitary U_{coll} and the distribution unitary U_{dist}.

- **Distribution Layer:** $U_{dist} = \prod CNOT_{m \to i}$. The CNOT gate is controlled on the messenger qubit (m) and targets the data qubit (i). Its Lie generators include terms proportional to $Z_m \otimes X_i$.
- **Data Register Layer:** U_{opr} applies local rotations and CZ gates strictly to the data register, contributing generators of the form $I_m \otimes A$, where $A \in \mathfrak{su}(2^n)$.

The capability of the circuit to saturate the extended $U(2^{n+1})$ Haar measure depends entirely on how the collection generators U_{coll} interact with $Z_m \otimes X_i$ and $I_m \otimes A$.

2 The R_{zz} Ablation: Algebraic Collapse

Suppose we replace the default R_{xx} collection gates with R_{zz} interactions. The corresponding generators are proportional to $Z_m \otimes Z_i$.

We evaluate the commutator between the collection and distribution generators:

$$[Z_m \otimes Z_i, Z_m \otimes X_i] = Z_m^2 \otimes [Z_i, X_i] = I_m \otimes (2iY_i) \tag{1}$$

Because Z_m commutes with itself, the messenger subspace trace of the commutator yields the identity I_m. Iterating Lie brackets between the set $\{I_m \otimes A, Z_m \otimes B\}$ strictly closes the algebra.

Theorem of Collapse: The DLA of the R_{zz} architecture cannot generate operators containing X_m or Y_m. The algebra fractures into $\mathfrak{g} \subset \mathfrak{su}(2^{n+1})$.

Physical Consequence: The messenger qubit is initialized in $|0\rangle_m$, an eigenstate of Z_m.

Because the DLA lacks X_m and Y_m generators, the messenger can never undergo population transfer. It remains permanently in $|0\rangle_m$. Consequently, the Z_m-controlled CNOT distribution layer evaluates trivially as the identity. The messenger acts as a fully decoupled classical wire, the active Hilbert space collapses to 2^n, and the ensemble variance advantage mathematically vanishes.

3 The R_{yy} Ablation: Algebraic Isomorphism

Suppose we instead substitute R_{yy} collection gates, yielding generators proportional to $Y_m \otimes Y_i$.

We evaluate the commutator against the distribution layer:

$$[Y_m \otimes Y_i, Z_m \otimes X_i] = (Y_m Z_m) \otimes (Y_i X_i) - (Z_m Y_m) \otimes (X_i Y_i) \tag{2}$$

Applying Pauli operator identities:

$$(iX_m) \otimes (-iZ_i) - (-iX_m) \otimes (iZ_i) = 2(X_m \otimes Z_i) \tag{3}$$

Because Y_m and Z_m do not commute, the Lie bracket successfully breaks out of the Z-eigenbasis, generating X_m. Subsequent commutations between $\{X_m \otimes A, Y_m \otimes B, Z_m \otimes C\}$ guarantee the generation of the full special unitary group.

4 Conclusion

The trainability advantage of the Q-LINK architecture is highly sensitive to the interaction basis of the residual pathway. The collection generator must not commute with the messenger-space basis of the distribution generator. While R_{yy} serves as a perfect theoretical substitute for R_{xx}, utilizing an R_{zz} basis destroys the architectural topology, collapsing the quantum residual connection into a disjointed scalar operation.